MATH19641: Mathematics 1Q1 Lecture Notes (Gappy version)

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Semester 1 – Academic Year 2026/27

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0 Introduction

Welcome to the course notes for Mathematics 1Q1! For a general introduction to the course, including details of assessments, please visit the Course Unit Information module on Canvas.

0.1 Help and support

Course unit lead: Carl Whitfield (you can call me Carl).
E-mail: carl.whitfield@manchester.ac.uk (Start with "Hi Carl").
Office hour: Mondays 10am-11am, Room 1.214 Alan Turing Building

Piazza: Ask questions (anonymously if you wish) and discuss topics with other students on the course in Piazza, accessed via the Canvas page for this course.

Student Support Hub at the Alan Turing Building:
Drop in for Advice, Guidance and Wellbeing Support, Monday to Friday 9:00-17:00

School of Natural Sciences Wellbeing Team: sons.wellbeing@manchester.ac.uk

University Student Support: Central university support and links to the counselling service.

Disability Advisory and Support Service: Register online or book an appointment to get a University Support Plan.

0.2 About the course

The aim of this course is to provide a comprehensive introduction to concepts in algebra and calculus for year 1 Chemistry students. Below you will find the Intended Learning Outcomes (ILOs) for this course unit.

Intended Learning Outcomes

On completion of this course unit, students will be able to

- **[ILO1]** Use equations in implicit or parametric form including straight lines or conic sections to extract relevant information.
- **[ILO2]** Identify the appropriate partial fractions form and find the values of any coefficients for a rational function of a single variable.
- **[ILO3]** Differentiate functions and hence locate and classify stationary points for a function of a single variable.
- **[ILO4]** Recall and apply appropriate rules to integrate functions and use these to measure areas associated with curves or combinations of curves.
- **[ILO5]** Calculate sums, differences and products involving vectors and use these to solve geometrical problems.
- **[ILO6]** Carry out arithmetic using complex numbers and hence solve polynomial and power equations.
- **[ILO7]** Model chemical processes using differential equation systems and solve these systems using a variety of techniques.

0.3 How to use these notes

These notes come in two forms,

- **[This version].** The "gappy notes", which contain gaps in the exercise solutions for you to fill as we work through the material in lectures
- The complete notes, which contains all the full solutions for examples in the lectures and supplementary videos (available on Canvas once the material has been covered in lectures).

In this PDF version of the notes, some of the HTML functionality is lost (such as interactive visualisations), to use these, please see the HTML version in Canvas.

The example exercises in this document are split into three types

- **[Lecture]** These examples will be covered in the lecture.
- **[DIY]** There will be time during lectures to complete these exercises.
- **[Video]** These examples are covered in videos, which can be accessed via Canvas.

0.4 AI use declaration

Generative AI (Microsoft Copilot) has been used to

- Transcribe the course slide deck into a first draft of these course notes, which were subsequently extensively edited manually.
- Generate the javascript scripts for interactive visualisations.
- Cross-check the final PDF version for errors or inconsistencies.

Otherwise, all course unit materials including the mathematical content, examples, diagrams, slides, tutorial questions, online quizzes, *etc.* were created by humans (*i.e.* the current and previous course unit leads).

5 Complex Numbers

Intended Learning Outcomes

The relevant ILOs for this section are

- **[ILO6]** Carry out arithmetic using complex numbers and hence solve polynomial and power equations.

5.1 Introduction to complex numbers

Learning Outcomes

On completion of this subsection, students will be able to

- Define real, imaginary and complex numbers and the complex conjugate
- Plot complex numbers on an Argand diagram

Consider the equation $z^2 - 8z + 15 = 0$. Completing the square

$$\begin{aligned}(z-4)^2 - 1 &= 0\\ z - 4 &= \pm 1\\ \therefore z &= 4 \pm 1\end{aligned}$$

and so $z = 3$ or $z = 5$.

Now, consider the equation $z^2 - 8z + 17 = 0$. Completing the square again

$$\begin{aligned}(z-4)^2 + 1 &= 0\\ z - 4 &= \pm\sqrt{-1}\\ \therefore z &= 4 \pm \sqrt{-1}\,.\end{aligned}$$

The solution does not exist, *i.e.* it has no 'real' roots. These two quadratics are plotted in figure 5.1.

Let's imagine that $\sqrt{-1}$ does exist, by which we mean that it is some quantity that we can manipulate algebraically. We define

$$i = \sqrt{-1}\,. \tag{5.1.1}$$

Here i is an "imaginary" number. You might also see this called j (but not in this course). Any multiple of i is also an imaginary number *e.g.* $3i$, $\sqrt{2}i$, $\frac{4i}{5}$, πi are all imaginary numbers.

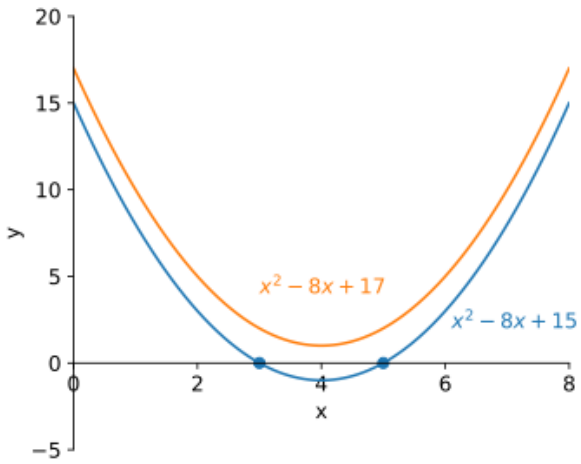


Figure 5.1: The blue line shows the example quadratic with two real roots, indicated by the dots. The orange line shows the example case with no real roots.

Example 5.1.1 (DIY)**.** Express the solution to the following equations in terms of i

$$z^2 = -1 \qquad w^2 = -16 \qquad v^2 = -28$$

Solution (left blank for your working)

Return to the equation $z^2 - 8z + 17 = 0$. We arrived at $z = 4 \pm \sqrt{-1}$ or $z = 4 \pm i$. This is a **complex number**. In **standard form**, a complex number is written as

$$z = x + yi \, . \tag{5.1.2}$$

It has a real part and an imaginary part $\mathrm{Re}\,(z) = x$ and $\mathrm{Im}\,(z) = y$. **Note:** The imaginary part is a real number!

The Argand diagram Figure 5.2 shows how a complex number can be understood as a two-

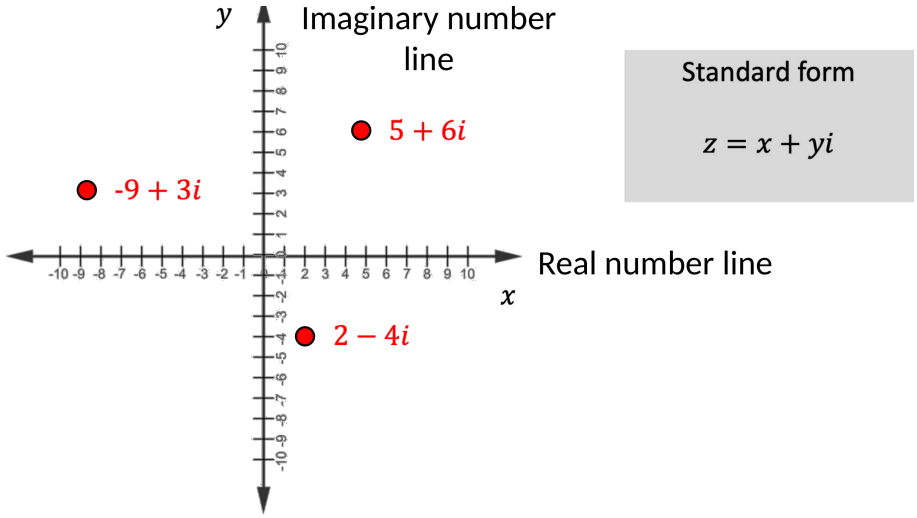


Figure 5.2: Complex numbers are comprised of two numbers (x and y in standard form), each of which can take any value on the number line.

dimensional number, consisting of a real part, which is on the real number line, and an imaginary part, which is on the imaginary number line.

Figure 5.3(a) shows an **Argand Diagram**. It plots the real part of a complex number on the horizontal axis and the imaginary part on the vertical axis.

The complex conjugate Every complex number $z = x + yi$ has its (complex) **conjugate** $\overline{z} = x - yi$. To find the conjugate, reverse the sign of the imaginary part. The conjugate is used in some operations e.g. division of complex numbers. It is also sometimes denoted z^*.

Example 5.1.2 (Lecture)**.** For the following complex numbers, find

(a) their real and imaginary parts

(b) their complex conjugates

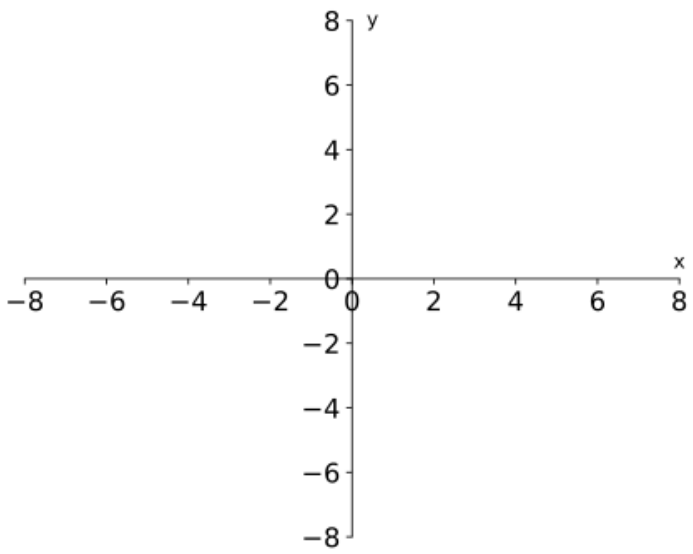
$$z_1 = 2 + 5i \qquad z_2 = -7 - 2i$$

Then plot $z_1, z_2, \overline{z}_1, \overline{z}_2$ on an Argand Diagram.

Solution

The real parts are $\mathrm{Re}\,(z_1) =$ _____ and $\mathrm{Re}\,(z_2) =$ _____. The imaginary parts are $\mathrm{Im}\,(z_1) =$ _____ and $\mathrm{Im}\,(z_2) =$ _____. The conjugates are $\overline{z}_1 =$ _____ and $\overline{z}_2 =$ _____.

Solution (cont)



Recap

Key points from section 5.1

- $i = \sqrt{-1}$ is an imaginary number (as is any multiple of i)
- A complex number is the sum of a real and imaginary number and can be written in standard form as $z = x + yi$
- An Argand Diagram plots the real part of a complex number on the x-axis and the imaginary part on the y-axis
- Taking the conjugate of a complex number means reversing the sign of the imaginary part, i.e. $\overline{z} = x - yi$

5.2 Arithmetic of complex numbers

Learning Outcomes

On completion of this subsection, students will be able to

- Add, subtract, multiply, and divide complex numbers in standard form.
- Find the magnitude of a complex number.

Addition and subtraction of complex numbers. For complex number arithmetic, we can treat i like a variable. Consider two complex numbers

$$\begin{aligned} z_1 &= x_1 + y_1 i \\ z_2 &= x_2 + y_2 i \ . \end{aligned}$$

These can be added together by adding the real parts and the imaginary parts separately

$$\begin{aligned} z_1 + z_2 &= (x_1 + y_1 i) + (x_2 + y_2 i) \\ &= (x_1 + x_2) + (y_1 + y_2) i \end{aligned}$$

The same is true for subtraction, subtract the real parts and imaginary parts separately

$$z_1 - z_2 = (x_1 - x_2) + (y_1 - y_2) i \ .$$

Multiplication and division of complex numbers. Consider two complex numbers $z_1 = x_1 + y_1 i$ and $z_2 = x_2 + y_2 i$. These can be multiplied by expansion

$$\begin{aligned} z_1 z_2 &= (x_1 + y_1 i)\,(x_2 + y_2 i) \\ &= x_1 x_2 + (x_1 y_2 + x_2 y_1)\, i + y_1 y_2 i^2 \\ &= (x_1 x_2 - y_1 y_2) + (x_1 y_2 + x_2 y_1) i \ , \end{aligned}$$

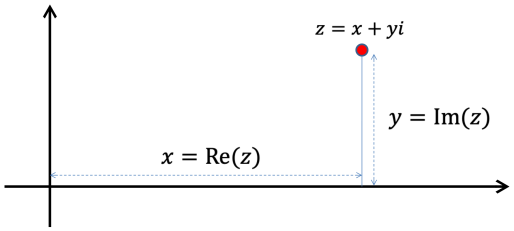
where in going from the second to the third line we used the definition $i^2 = -1$.

The **modulus** or **magnitude** of a complex number is $|z| = \sqrt{z\bar{z}}$ Defining $z = x + yi$

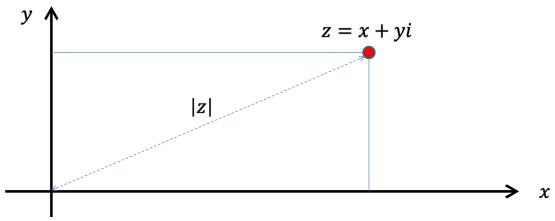
$$\begin{aligned} |z| &= \sqrt{(x + yi)(x - yi)} \\ &= \sqrt{x^2 + y^2} \ . \end{aligned} \tag{5.2.1}$$

Therefore, $|z|$ is the distance from the origin on the Argand diagram, as shown in figure 5.3(b).

Consider again the two complex numbers $z_1 = x_1 + y_1 i$ and $z_2 = x_2 + y_2 i$, the quotient is also a



(a) Re(z) and Im(z).



(b) $|z|$.

Figure 5.3: Two plots of a complex number on an Argand diagram. (a) The real part is the x-coordinate and the imaginary part is the y-coordinate. (b) The modulus is the distance from the origin.

complex number

$$\frac{z_1}{z_2} = \frac{x_1 + y_1 i}{x_2 + y_2 i}\,,$$

however it is not immediately clear how to write it in standard form. Compare this to dividing z_1 by a real number a

$$\frac{z_1}{a} = \frac{x_1 + y_1 i}{a} = \frac{x_1}{a} + \frac{y_1}{a} i\,.$$

In this case it is straightforward to write this in standard form.

In the case of z_1/z_2, we can always turn the complex number in the denominator z_2 into a real number by multiplying top and bottom by $\overline{z_2}$

$$\frac{z_1}{z_2} = \frac{z_1\overline{z_2}}{z_2\overline{z_2}} = \frac{z_1\overline{z_2}}{|z_2|^2}\,. \tag{5.2.2}$$

We can now use the rules of complex number multiplication to write this quotient in standard form

$$\frac{z_1}{z_2} = \frac{(x_1 + y_1 i)(x_2 - y_2 i)}{(x_2)^2 + (y_2)^2} = \frac{x_1 x_2 + y_1 y_2}{(x_2)^2 + (y_2)^2} + \frac{x_2 y_1 - x_1 y_2}{(x_2)^2 + (y_2)^2} i\,.$$

Example 5.2.1 (Lecture)**.** Given the two complex numbers $z_1 = -3 + 4i$ $z_2 = 5 - 5i$ find
(a) $z_1 + z_2$ and $z_1 - z_2$ (b) $z_1 z_2$ (c) $\frac{z_1}{z_2}$

Solution

(a) $z_1 + z_2 =$ $=$

$z_1 - z_2 =$ $=$

(b) $z_1 z_2 =$

$=$ $=$.

(c) $\frac{z_1}{z_2} = \frac{(-3+4i)}{(5-5i)} =$

$=$ $=$

Example 5.2.2 (Video)**.** Find the complex values of z that satisfy the equation

$$z\overline{z} + 11z + 9\overline{z} + 3 - 8i = 0$$

Solution

First write $z = x + yi$, such that $\overline{z} =$ and sub. in

$= 0$

$\therefore$ $= 0$

Taking the imaginary part of both sides gives so $y =$.

Taking the real part of both sides gives or

Solution (cont)

$\square = 0$, which factorises to

$$\square = 0$$

and so $x = \square$ or $x = \square$. Putting it all together this means $z = -\square$ or $z = \square$

Recap

Key points from section 5.2

- The imaginary number i can be treated like a variable where $i^2 = -1$
- To add complex numbers, simply add their real and imaginary parts.
- To multiply, expand out and collect the real and imaginary parts.
- To divide, multiply both numerator and denominator by the conjugate of the denominator, so that the denominator becomes a real number.
- The magnitude of a complex number $z = x + yi$ is $|z| = \sqrt{x^2 + y^2}$.

5.3 Polar and Exponential Forms

Learning Outcomes

On completion of this subsection, students will be able to

- Convert complex numbers between standard, polar, and exponential forms
- Perform multiplication, division and power operations with complex numbers in polar and exponential form

Consider a point (a complex number) on an Argand diagram. We can convert this point to polar coordinates (see figure 5.4 using the relations $x = r\cos\theta$, $y = r\sin\theta$, $r = \sqrt{x^2+y^2}$, and $\tan\theta = \frac{y}{x}$. Thus, the complex number $z = x + yi$ can be written in polar form as

$$z = x + yi = r(\cos\theta + i\sin\theta)\,, \tag{5.3.1}$$

where $r = |z|$ is the modulus or magnitude of the complex number $\theta = \text{Arg}(z)$ (normally $-\pi \leq \theta \leq \pi$ or $0 \leq \theta \leq 2\pi$) is the argument of the complex number.

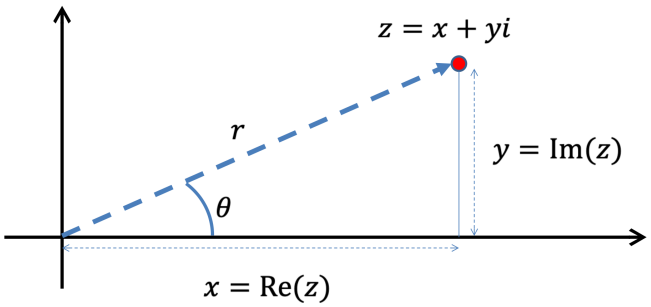


Figure 5.4: An Argand diagram where the complex number z is given by its polar coordinates r and θ.

Example 5.3.1 (DIY)**.** Write the following complex numbers in standard form.
(a) z_1 which has modulus 5 and argument $\frac{\pi}{3}$
(b) z_2 where $|z_2| = 3$ and $\text{Arg}\,(z_2) = \frac{7\pi}{6}$

Solution (left blank for your working)

Example 5.3.2 (Lecture)**.** Convert the following complex numbers from standard to polar form.
(a) $z_1 = 2 + 2i$ (b) $z_2 = -\sqrt{3} - i$

Solution

(a) Write $z_1 = r_1 (\cos\theta_1 + i \sin\theta_1)$ then $r_1 =$.

z_1 is in the quadrant, so $\theta_1 =$ therefore

$z_1 =$ for $0 \leq \theta \leq 2\pi$.

Solution (cont)

(b) Write $z_2 = r_2\left(\cos\theta_2 + i\sin\theta_2\right)$ then $r_2 =$.

z_2 is in the quadrant, so $\theta_2 =$, therefore

$$z_2 = \qquad \text{for } 0 \le \theta \le 2\pi \ .$$

Multiplication and Division in Polar Form. Consider the two complex numbers $z_1 = r_1\left(\cos\theta_1 + i\sin\theta_1\right)$ $z_2 = r_2\left(\cos\theta_2 + i\sin\theta_2\right)$. Multiplying them together gives

$$\begin{aligned} z_1 z_2 &= r_1 r_2\left(\cos\theta_1 + i\sin\theta_1\right)\left(\cos\theta_2 + i\sin\theta_2\right) \\ &= r_1 r_2\left(\cos\theta_1\cos\theta_2 + i\sin\theta_1\cos\theta_2 + i\cos\theta_1\sin\theta_2 + i^2\sin\theta_1\sin\theta_2\right) \\ &= r_1 r_2\left[\left(\cos\theta_1\cos\theta_2 - \sin\theta_1\sin\theta_2\right) + i\left(\sin\theta_1\cos\theta_2 + \cos\theta_1\sin\theta_2\right)\right] \\ &= r_1 r_2\left[\cos\left(\theta_1+\theta_2\right) + i\sin\left(\theta_1+\theta_2\right)\right] \ . \end{aligned}$$

Therefore,

$$z_1 z_2 = r_1 r_2\left[\cos\left(\theta_1+\theta_2\right) + i\sin\left(\theta_1+\theta_2\right)\right] \ , \tag{5.3.2}$$

and

$$\frac{z_1}{z_2} = \frac{r_1}{r_2}\left[\cos\left(\theta_1-\theta_2\right) + i\sin\left(\theta_1-\theta_2\right)\right] \ . \tag{5.3.3}$$

I.e. to multiply two complex numbers, multiply their moduli and add their arguments. To divide them, divide their moduli and subtract the argument of the denominator from the argument of the numerator.

Example 5.3.3 (Lecture)**.** Convert $z_1 = -2 + 2\sqrt{3}i$ and $z_2 = 5 - 5i$ to polar form. Hence, find $z_1 z_2$ and $\frac{z_1}{z_2}$.

Solution

Write $z_1 = r_1\left(\cos\theta_1 + i\sin\theta_1\right)$ and $z_2 = r_2\left(\cos\theta_2 + i\sin\theta_2\right)$ where $r_1 =$

and $r_2 =$.

Solution (cont)

The argument θ_1 is in the second quadrant, so $\theta_1 =$. The argument θ_2 is in the fourth quadrant, so $\theta_1 =$ or .

So,

$z_1 =$ and $z_2 =$

Then, multiplying the moduli and adding the arguments we find

$z_1 z_2 = = .$

Dividing the moduli and subtracting the arguments we find

$\frac{z_1}{z_2} = = .$

Powers in Polar Form. It follows from equation (5.3.2) that if we raise the complex number $z = r(\cos\theta + i\sin\theta)$ to the power n, we get

$$z^n = r^n \left[\cos(n\theta) + i\sin(n\theta)\right] . \tag{5.3.4}$$

I.e. raise the modulus to the power n and multiply the argument by n. This is known as **De Moivre's theorem**. This is visualised in the animation in figure 5.5.

The Exponential Form. Note that this polar form can be written in an alternative manner

$$r(\cos\theta + i\sin\theta) \equiv re^{i\theta} . \tag{5.3.5}$$

This is known as **Euler's formula**. For the special case $r = 1$ and $\theta = \pi$, we see

$$e^{i\pi} = -1 ,$$

Also known as Euler's identity, often written $e^{i\pi} + 1 = 0$. This equation elegantly links some of the most fundamental mathematical constants e, i, π, and -1.

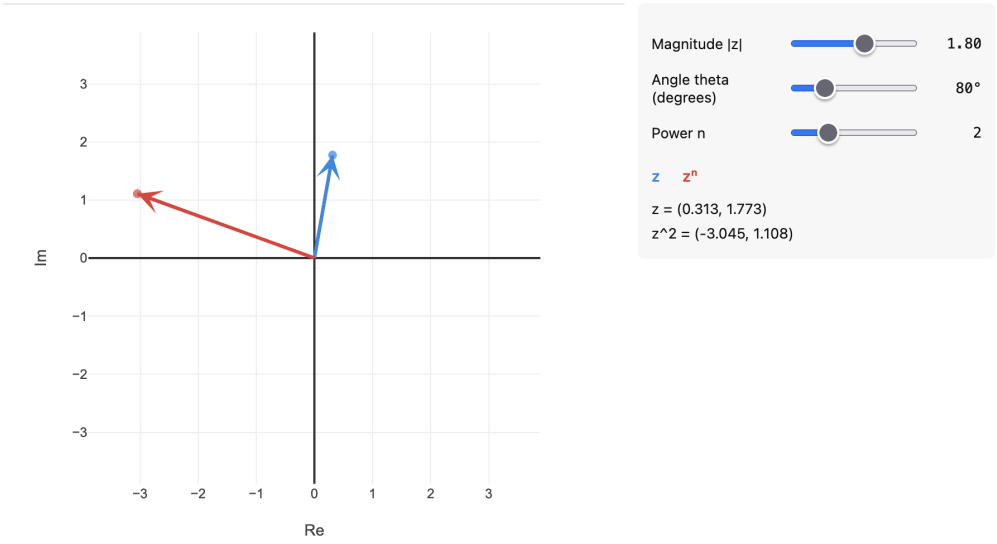


Figure 5.5: Screenshot of the interactive plot of the complex numbers z (blue) and z^n (red) on an Argand diagram (available in the HTML version of these notes).

It follows that, if $z_1 = r_1 e^{i\theta_1}$ and $z_2 = r_2 e^{i\theta_2}$ then

$$z_1 z_2 = r_1 r_2 e^{i(\theta_1+\theta_2)}$$
$$\frac{z_1}{z_2} = \frac{r_1}{r_2} e^{i(\theta_1-\theta_2)}$$
$$z_1^n = r_1^n e^{in\theta_1}$$

Therefore, under these basic arithmetic operations, the complex exponential behaves the same as the exponential of real numbers.

Example 5.3.4 (DIY)**.** Find

$$\left(-\frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}}\right)^{17}$$

and express in the form $x + yi$.

Solution (left blank for your working)

Example 5.3.5 (Video)**.** Use De Moivre's theorem to find an identity for $\cos 4\theta$ in terms of $\cos\theta$ and $\sin\theta$.

Solution

Defining $z = (\cos\theta + i\sin\theta)$, then from De Moivre's theorem for $n = 4$

$$z^4 = [\cos(4\theta) + i\sin(4\theta)]$$

Using direct multiplication

$$z^2 = z \times z = \qquad \equiv$$

and again

$$z^4 = z^2 \times z^2 =$$

Comparing real parts of both sides $\mathrm{Re}\,(z^4) = a^2 - b^2$. Then, using the expression from De

Solution (cont)

Moivre's theorem

$$\cos(4\theta) =$$

$$\therefore \cos(4\theta) =$$

Recap

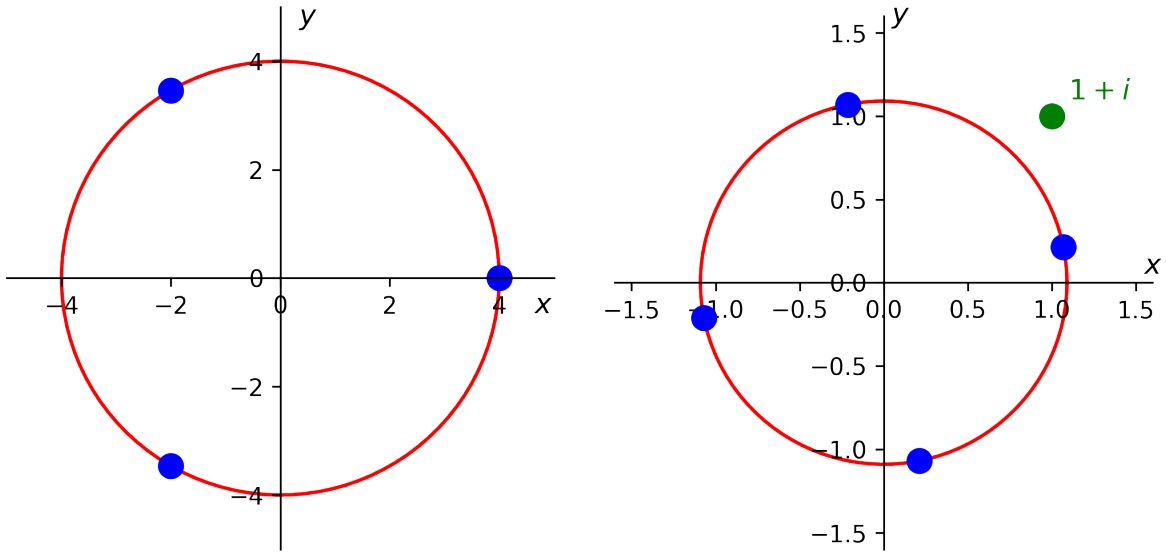
Key points from section 5.3

- Polar and exponential forms express a complex number in terms of its modulus and argument

$$x + iy \equiv r(\cos\theta + i\sin\theta) \equiv re^{i\theta}$$

- Multiplication, division and powers of complex numbers are simpler to calculate in these forms e.g.

$$\begin{aligned} z_1 z_2 &= r_1 r_2 e^{i(\theta_1+\theta_2)} \frac{z_1}{z_2} \\ &= \frac{r_1}{r_2} e^{i(\theta_1-\theta_2)} z_1^n \\ &= r_1^n e^{in\theta_1} \end{aligned}$$



(a) (b)

Figure 5.6: (a) The roots of the equation $z^3 = 64$ on an Argand diagram are equally spaced around the circle of radius 4 centred at the origin. (b) The roots of the equation $z^4 = 1 + i$ on an Argand diagram.

5.4 Complex Roots

Learning Outcomes

On completion of this subsection, students will be able to

- Find all the complex roots of a monomial or polynomial equation
- Plot the roots of a complex number on an Argand diagram

Roots of Complex Numbers. Consider the equation

$$z^2 = 64$$

This equation has two solutions i.e. $z = 8$ and $z = -8$. Now consider the equation

$$z^3 = 64$$

One solution is $z = 4$, but it is a cubic so there are two more solutions $z = -2 + 2\sqrt{3}i$ and $z = -2 - 2\sqrt{3}i$.

The solutions to $z^3 = 64$ are equally spaced around a circle. $z_1 = 4$, $z_2 = -2 + 2\sqrt{3}i$ and $z_3 = -2 - 2\sqrt{3}i$, see figure 5.6(a).

In general, the equation

$$z^n = a + bi$$

will have n solutions that are spaced equally around a circle in the Argand Diagram. For example, the roots of the equation $z^4 = 1 + i$ are plotted in 5.6(b).

To find the solutions of $z^n = a + bi$ first express $a + bi$ in exponential form *i.e.*

$$a + bi = re^{i\theta} .$$

Now, note that 2π can be added to the argument without changing the complex number itself $a + bi = re^{i\theta} = re^{i(\theta+2\pi)} = re^{i(\theta+4\pi)} = \ldots = re^{i(\theta+2(n-1)\pi)}$. This gives n different, but equivalent arguments. Dividing these by n gives all the roots for $0 \leq \theta \leq 2\pi$.

Returning to $z^3 = 64$, we can rewrite as $z^3 = 64e^{0i}$ and furthermore as

$$z^3 = 64e^{0i} = 64e^{2\pi i} = 64e^{4\pi i} .$$

Taking the cube root, the modulus is the real root of $\sqrt[3]{64} = 4$, and the argument is divided by 3, resulting in the three solutions

$$z_1 = 4e^{0i} \qquad z_2 = 4e^{\frac{2\pi i}{3}} \qquad z_3 = 4e^{\frac{4\pi i}{3}} .$$

Converting back to standard form

$$\begin{aligned}
z_1 &= 4 \\
z_2 &= 4\left[\cos\left(\frac{2\pi}{3}\right) + i\sin\left(\frac{2\pi}{3}\right)\right] \\
&= 4\left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i\right) = -2 + 2\sqrt{3}i \\
z_3 &= 4\left[\cos\left(\frac{4\pi}{3}\right) + i\sin\left(\frac{4\pi}{3}\right)\right] \\
&= 4\left(-\frac{1}{2} - \frac{\sqrt{3}}{2}i\right) = -2 - 2\sqrt{3}i
\end{aligned}$$

As shown in figure 5.6, the solutions are equally spaced around a circle. $z_1 = 4$, $z_2 = -2 + 2\sqrt{3}i$ and $z_3 = -2 - 2\sqrt{3}i$ $z_1 = 4e^{i0}$, $z_2 = 4e^{\frac{i2\pi}{3}}$ and $z_3 = 4e^{-\frac{i2\pi}{3}}$.

Example 5.4.1 (Lecture)**.** Find all solutions to the equation

$$z^4 = -9$$

Give your answers in the form $z = a + bi$.

Solution

In exponential form $-9 =$, incrementing the argument by 2π

$$z^4 = \quad = \quad = \quad = \quad$$

The modulus of the roots is and so the roots are

$$z_1 = , \qquad z_2 = ,$$

$$z_3 = , \qquad z_4 = .$$

Converting to standard form

$$z_1 = \quad = ,$$

$$z_2 = \quad = ,$$

$$z_3 = \quad = ,$$

$$z_4 = \quad = .$$

Roots of Polynomial Equations. An n-th degree polynomial equation can have up to n real roots, some of which may be repeated. Including complex roots, an n-th degree polynomial equation will always have n solutions (including repeated roots). *E.g.*

$$z^3 + z^2 - 7z - 15 = 0$$

has solutions $z = 3$, $z = -2 + i$, and $z = -2 - i$.

Recall the example with no real roots from the start of this section

$$f(z) = z^2 - 8z + 17 = 0\,.$$

The complex roots are where both the real and imaginary parts of $f(z)$ are 0 (see figure 5.7).

If $z = a + bi$ is a solution to a polynomial equation with real coefficients, then so is the conjugate

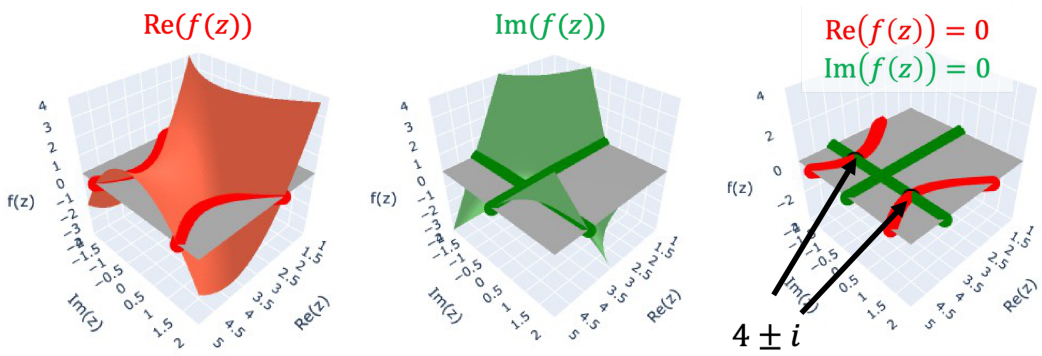


Figure 5.7: Visualisation of the complex roots of $f(z) \equiv z^2 - 8z + 17 = 0$ where the x and y axes are the real and imaginary parts of z respectively and the grey plane corresponds to $f(z) = 0$. Left: The red surface shows the real value of $f(z)$. Middle: The green surface shows the imaginary values of $f(z)$. Right: The green and red lines show where the real and imaginary parts of $f(z)$ are zero, respectively. Where these lines cross are the roots of the equation.

$\overline{z} = a - bi$. It follows that $z - a - bi$ and $z - a + bi$ will be factors of the polynomial and so

$$\begin{aligned}(z - a - bi)\,(z - a + bi) &= (z - a)^2 + b^2 \\ &= z^2 - 2az + a^2 + b^2\,,\end{aligned}$$

is a factor of the polynomial.

Example 5.4.2 (Lecture)**.** Given that $z = 1 + i$ is a solution to the equation

$$z^3 + 2z^2 - 6z + 8 = 0$$

find the other solutions.

Solution

The cubic equation has real coefficients, so $z =$ [ ] must also be a solution. Thus, we can factorise the cubic as

$$\begin{aligned}z^3 + 2z^2 - 6z + 8 &= \\ &= \\ &=\end{aligned}$$

Comparing coefficients, we see the final root is $z =$ [ ].

Example 5.4.3 (Video)**.** Find the complex values of z that satisfy the equation

$$z^2 + (2+4i)\,z - 3 + \frac{7}{4}i = 0$$

Solution

Use the quadratic equation (or discriminant method)

$$z = -\frac{b}{2} + \frac{1}{2}\left(b^2 - 4c\right)^{\frac{1}{2}}$$

with $b =$ and $c = -$. This gives

$z =$

$=$

Let $w =$, in exponential form $w =$ $=$. Then

$w^{\frac{1}{2}} =$ $=$

or

$w^{\frac{1}{2}} =$ $=$.

So, $z =$

or

$z =$.

Recap

Key points from section 5.4

- A complex (or real) number to the power $1/n$ has n roots
- To find the roots, generate n equivalent complex numbers by adding 2π to the argument

Recap (cont)

$n-1$ times $a+bi = re^{i\theta} = re^{i(\theta+2\pi)} = re^{i(\theta+4\pi)} = \ldots = re^{i(\theta+2(n-1)\pi)}$

- An n-th order polynomial also has n roots (some may be repeated)
- If the polynomial has real coefficients, then any complex roots $z = a + bi$ will occur in a pair with its conjugate $z = a - bi$

5.5 Complex Numbers and Loci

Learning Outcomes

On completion of this subsection, students will be able to

- Sketch loci in an Argand diagram based on equations or inequalities involving complex numbers.

An equation (or inequality) involving a complex number $z = x + yi$ can be an efficient way to represent a line, curve or area in x-y space. Writing in terms of x and y can help to sketch the locus. A reminder of some relationships are given in table 1.

Complex representation	x-y representation
$\lvert z\rvert$	$\sqrt{x^2+y^2}$
$\mathrm{Re}\,(z)$	x
$\mathrm{Im}\,(z)$	y
$\lvert z-w\rvert$	$\sqrt{(x-a)^2+(y-b)^2}$
$\mathrm{Re}\,(z-w)$	$x-a$
$\mathrm{Im}\,(z-w)$	$y-b$

Table 1: A list of equivalent expressions in terms of the complex numbers $z = x+yi$ and $w = a+bi$ and in terms x, y, a, and b. Note: All of these expressions are real, they do not contain i.

Consider the expression $|z-w|$ which is equivalent to $\sqrt{(x-a)^2+(y-b)^2}$, and so can describe loci associated with the circle with centre at $w = (a, b)$. Some relations involving this expression are sketched in figure 5.8.

For example, consider the locus given by

$$2 < |z| \leq 5$$

It includes all points that are outside the circle $x^2+y^2 = 4$ and inside or on the circle $x^2+y^2 = 25$. This is shown in figure 5.9.

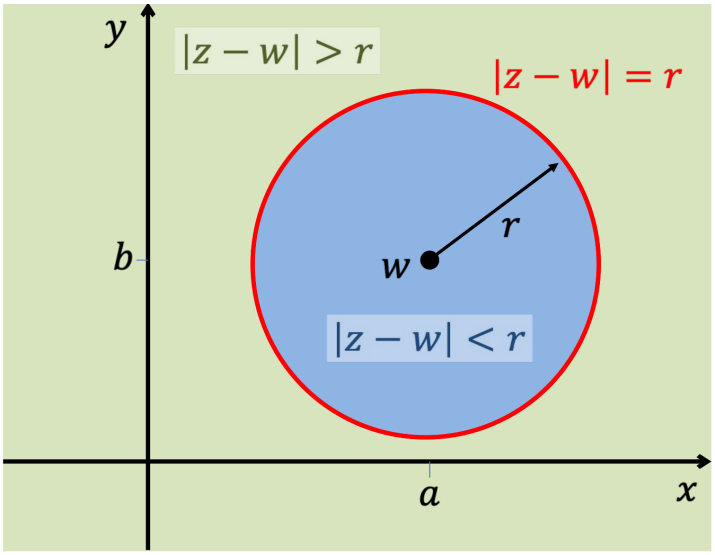


Figure 5.8: A sketch of three loci involving the circle $|z-w|=r$. When this is an inequality, the locus is all the points inside or outside the circle (blue and green). When it is an equality the locus contains all the points on the circle (red).

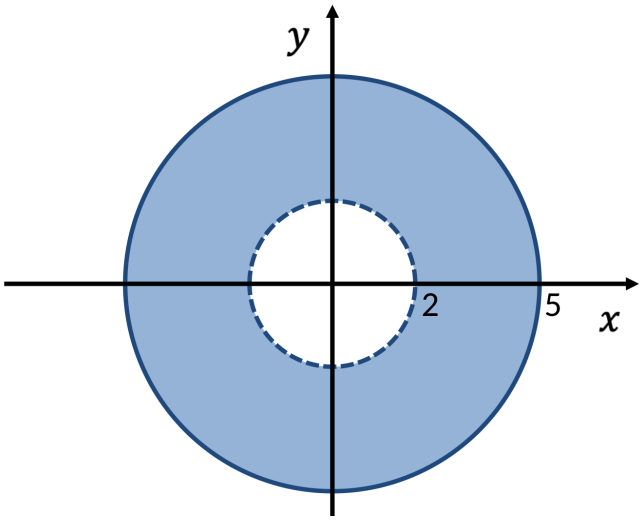


Figure 5.9: The shaded blue region indicates the locus $2 < |z| \leq 5$.

Example 5.5.1 (Lecture)**.** Find and sketch the loci described by

(a) $2\,\mathrm{Re}\,(z) + \mathrm{Im}\,(z - (1+2i)) = 10$

(b) $|z| \leq \mathrm{Re}\,(z+2)$

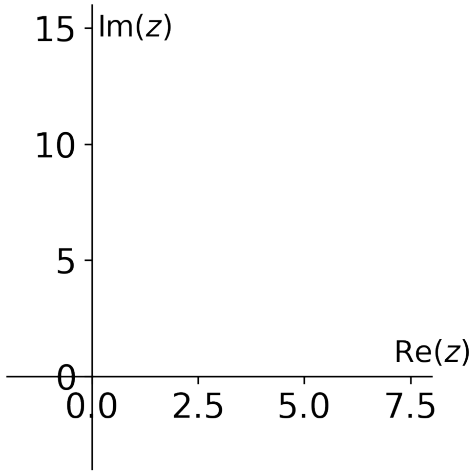
(c) $|z - 2 + i| < 3$

Solution

Express the loci in terms of x and y.

(a) $\mathrm{Re}\,(z) =$ [ ] and $\mathrm{Im}\,(z - (1+2i)) =$ [ ] so, the locus is

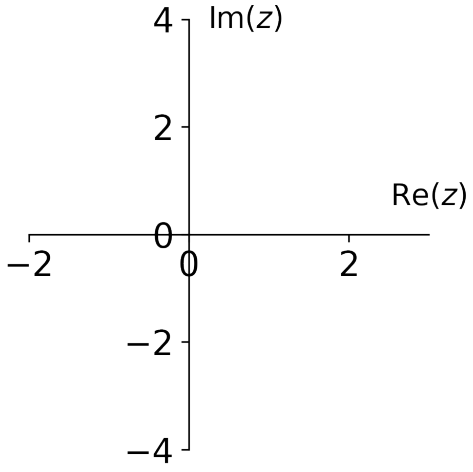
[ ] or [ ].



(b) $|z| =$ [ ] and $\mathrm{Re}\,(z+2) =$ [ ] so the locus is [ ] or [ ] which can be written

[ ].

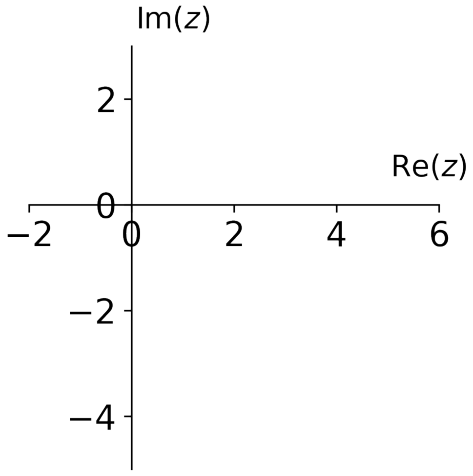
Solution (cont)



(c) $|z-2+i|$ = so the locus is or equivalently

.

I.e. all the points inside a circle with radius and centre .

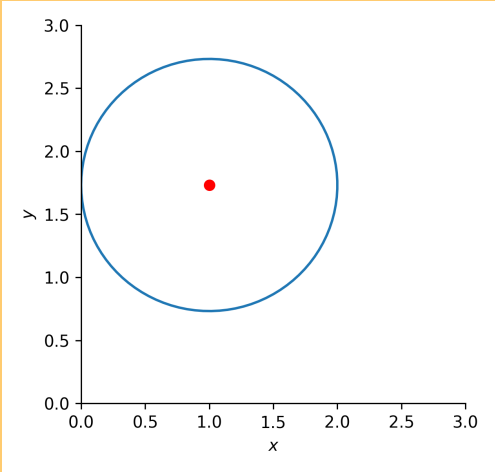


⋆ Exam mistake of the week ⋆

Complex loci may not be intuitive at first and so it is difficult to "read off" what the graph will be.

> **25/26 Exam Q5(a)(iii)** Given $w = 1 + \sqrt{3}i$ ...sketch the locus $|z - w| = 1$ on an Argand diagram.

Answer: A circle of radius 1 with centre w.



Many said it was a straight line, several others noticed it was a circle but put the centre at (0,0).
→ The best approach is to check by substituting in $z = x + yi$ and finding the Cartesian form.

Recap

Key points from section 5.5

- Equations and inequalities using complex numbers can be an efficient way to represent sets of points (loci) in x-y space.
- These could be points, lines, or areas depending on the form of the expression, but the resulting expression should not contain i.
- Writing the expression in terms of x and y can help to find the curves
- Some simple examples are the equation of a circle $|z - w| = R$ and straight line $\text{Im}(z) = mRe(z + c)$ (where m is real)